

ELECENG 2E14: Electronic Devices and Circuits 1

Design Project 2

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As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario. Submitted by [Anushka Chauhan, Chauha30 ,400598874]

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Ideal Switches Properties

- **Short Circuit when ON**

When an ideal switch is in the ON state, it behaves as a perfect short circuit. This means its resistance is exactly $0\ \Omega$, and the voltage across the switch is $0\ \text{V}$ regardless of the amount of current flowing through it. Because there is no voltage drop, no power is dissipated in the switch during conduction ($P = VI = 0$). The switch can carry any magnitude of current without overheating, degradation, or limitation [1], [2].

- **Open Circuit when OFF**

When the ideal switch is in the OFF state, it behaves as a perfect open circuit. Its resistance becomes infinite, which means the current through the switch is exactly $0\ \text{A}$. It completely blocks current flow with no leakage. Even if a voltage is applied across it, no current will pass through the device. Since current is zero, the switch dissipates no power in the OFF state [1], [2].

- **Works for any voltage or current**

An ideal switch has unlimited voltage and current capability. When ON, it can conduct any amount of current without restriction. When OFF, it can withstand any magnitude of voltage without breakdown or failure. There are no maximum ratings, thermal limits, or safe operating area constraints. This assumption simplifies circuit analysis because the switch operation is independent of the applied voltage levels v_1 or v_2 [1], [2].

- **Bidirectional Operation**

An ideal switch is bidirectional in both voltage blocking and current conduction. When ON, it allows current to flow in either direction with zero resistance. When OFF, it blocks the voltage of either polarity. This means the switch's behaviour does not depend on the direction of current or the polarity of the applied voltage, making it symmetrical in operation [1], [2].

Switch Non-idealities

i. Qualitative Description of Non-Idealities in a Real Switch

1. Non-zero ON resistance ($R_{on} > 0$)

Unlike an ideal switch, a real switch does not behave as a perfect short circuit when ON. It has a small but finite ON resistance. This resistance causes a voltage drop across the device during conduction and results in power dissipation as heat. Therefore, the switch is not lossless in the ON state [1], [2].

2. Non-zero OFF current ($I_{off} > 0$)

A real switch does not act as a perfect open circuit when OFF. Due to internal semiconductor properties, a small leakage current flows even in the OFF state. This leakage means the switch cannot completely block current, and a small power loss occurs even when it is supposed to be open [1], [2], [3].

3. Limited Voltage Range ($V_{min} < V_1, V_2 < V_{max}$)

Real switches have minimum and maximum voltage ratings. They can only operate safely within a specific voltage range. If the applied voltage exceeds the maximum rated voltage V_{max} , breakdown or permanent damage may occur. This means real switches cannot block unlimited voltage like ideal switches [1], [2].

4. Practical bidirectional behaviour (must be tested in both directions)

Although many practical switches are designed to conduct in both directions when ON and block voltage of either polarity when OFF, their characteristics may not be perfectly symmetrical. The ON resistance and leakage current can differ depending on the current direction. Therefore, functionality and resistance must be tested in both forward and reverse directions [1], [2].

ii. Quantitative Description of Non-Idealities in a Real Switch

1. Effect of non-zero ON resistance ($R_{on} > 0$)

Because a real switch has a finite ON resistance, a voltage drop appears across it when current flows. The voltage across the switch in the ON state is:

$$V_{ON} = I \times R_{on}$$

The conduction power loss is:

$$P_{ON} = I^2 \times R_{on}$$

If the switch operates with a duty cycle d , the average conduction loss becomes:

$$P_{ON(avg)} = I^2 \times R_{on} \times d$$

This power is dissipated as heat and reduces efficiency [1], [2].

2. Effect of OFF-state leakage current ($I_{off} > 0$)

Since a real switch has finite OFF resistance, a small leakage current flows even in the OFF state. The power dissipated due to leakage is:

$$P_{OFF} = V \times I_{off}$$

Although usually small, this represents non-zero energy loss [1], [2].

3. Effect of limited voltage range

A real switch must operate within its rated voltage limits:

$$V_{min} < V_1, V_2 < V_{max}$$

If the applied voltage exceeds V_{max} , breakdown may occur, leading to device failure.

These limits define the safe operating region of the switch [1], [2].

4. Quantifying bidirectional behaviour

The ON resistance must be measured in both directions to verify symmetry:

$$R_{on(forward)} = V_f / I_f$$

$$R_{on(reverse)} = V_r / I_r$$

In practical devices:

$$R_{on(forward)} \neq R_{on(reverse)}$$

Similarly, leakage current may differ depending on voltage polarity [1], [2], [3].

Test Plan

The purpose of this test plan is to quantitatively evaluate how closely the designed switches approximate ideal voltage-controlled switches. The four key performance metrics evaluated are:

1. ON-state resistance ($R_{on} > 0$)
2. OFF-state leakage current ($I_{off} > 0$)
3. Maximum functional voltage range
4. Bidirectional operation

Switch 1

All tests are performed under the design constraint:

- $V_{supply} = 5\text{ V}$
- $V_1 = 5\text{ V}$
- $V_{control} = 0\text{--}5\text{ V}$ square waveform
- R_L connected from V_2 to ground

The following table summarizes the Test plan for switch 1:

Test for ON-State Resistance (R_{on}) Ideal behaviour: - $V_1 \approx V_2$ - R_{on} is as small as possible	Values Set • $V_{supply} = 5\text{ V}$ • $V_{control} = 0\text{ V}$ (switch ON) • $V_1 = 5\text{ V}$ • Load resistor R_L connected from V_2 to ground	Values Measured • Voltage at V_2 • Current through $R_L(I)$	Values Calculated • On Resistance $R_{on} = (V_1 - V_2) / I$
Test for OFF-State Leakage Current (I_{off}) Ideal behaviour: - $I_{off} \approx 0$	Values Set • $V_{supply} = 5\text{ V}$ • $V_{control} = +5\text{ V}$ (switch OFF) • $V_1 = 5\text{ V}$	Values Measured • Voltage at V_2 • Current through $R_L(I)$	Values Calculated • Leakage Current I_{off}
Test for Maximum Functional Voltage	Values Set • $V_1 = 5\text{ V}$ • $V_{control}$ toggles	Values Measured • V_2 during ON and OFF	Values Calculated • $V_1 - V_2$ remains small when ON

	0 to 5 V		• I_{off} remains acceptably small when OFF
Test for Bidirectional Operation	Values Set • $V_1 = 5V$ • $V_{control}$ toggles $0 \leftrightarrow 5V$	Values Measured • R_{on} in forward direction • R_{on} in reverse direction	Values Calculated • $R_{on}(forward) = V_f / I_f$ • $R_{on}(reverse) = V_r / I_r$

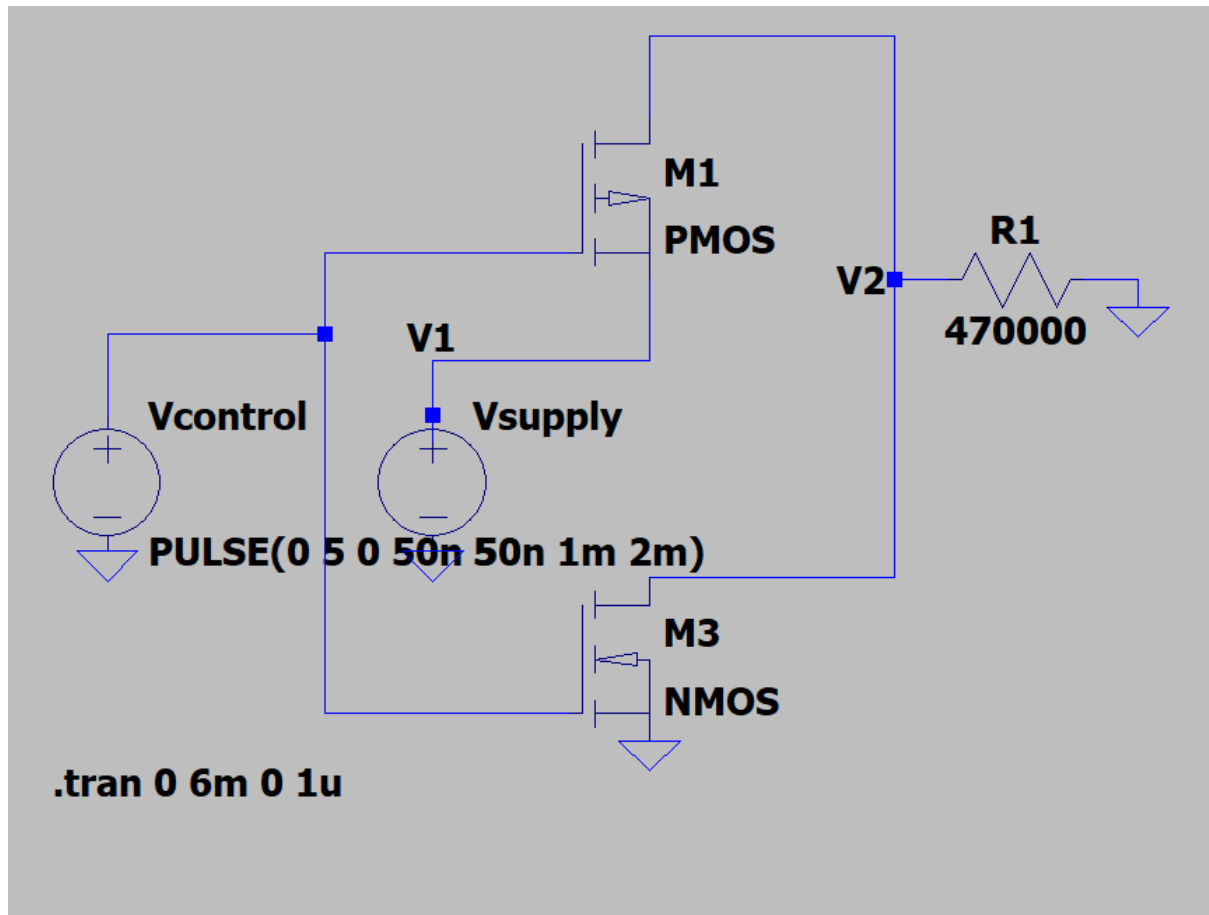
Switch 2

The following table summarizes the Test plan for switch 2:

Test for ON-State Resistance (R_{on})	Values Set • $V_{supply} = 5V$ • $V_{control} = 0V$ (switch ON) • Load Resistors: $470k\Omega$	Values Measured • Voltage at V_a • Voltage at V_b • Current through both resistors	Values Calculated • On Resistance $R_{on} = (V_1 - V_2) / I$
Test for OFF-State Leakage Current (I_{off})	Values Set • $V_{supply} = 5V$ • $V_{control} = +5V$ (switch OFF) • Load Resistors: $470k\Omega$	Values Measured • Voltage at V_a • Voltage at V_b • Current through both resistors	Values Calculated • Leakage Current I_{off}
Test for Maximum Functional Voltage	Values Set • V_{supply} and input = $5V$ • $V_{control}$ toggles $0 \leftrightarrow 5V$	Values Measured • V_a and V_b during ON and OFF	Values Calculated • When control = $0V \rightarrow V_a$ and V_b follow the input • When control = $5V \rightarrow V_a$ and $V_b \approx 0V$
Test for Bidirectional Operation	Values Set • Apply $5V$ at V_a and observe V_b • Apply $5V$ at V_b and observe V_a	Values Measured • R_{on} (V_a to V_b) • R_{on} (V_b to V_a)	Values Calculated • R_{on}

Switch Type 1

Circuit schematic and Netlist



*

M1 V2 N001 V1 V1 PMOS

M3 V2 N001 0 0 NMOS

R1 0 V2 470000

Vsupply V1 0 5

Vcontrol N001 0 PULSE(0 5 0 50n 50n 1m 2m)

.model NMOS NMOS

.model PMOS PMOS

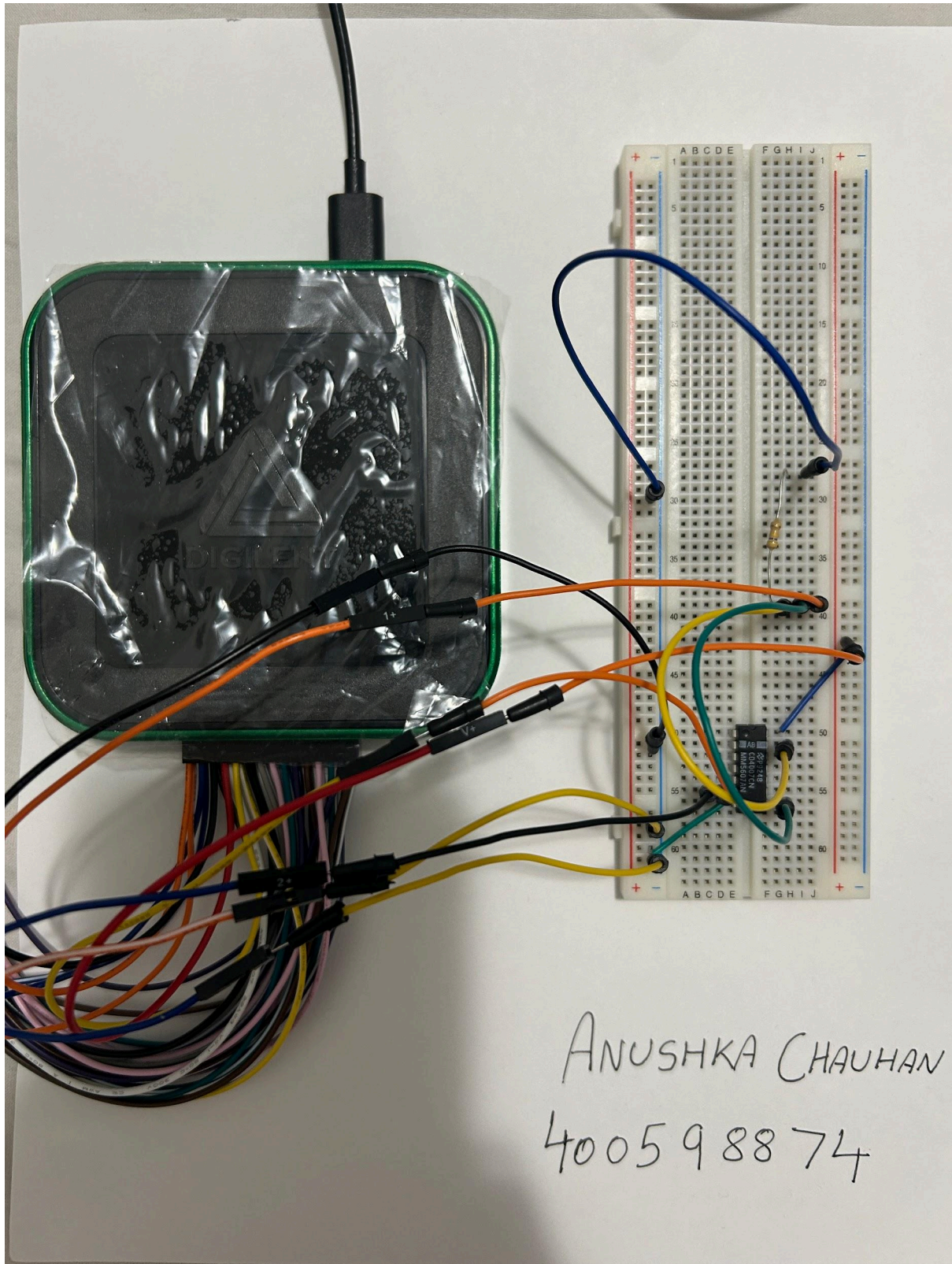
.lib C:\Users\camit\AppData\Local\LTspice\lib\cmp\standard.mos

.tran 0 6m 0 1u

.backanno

.end

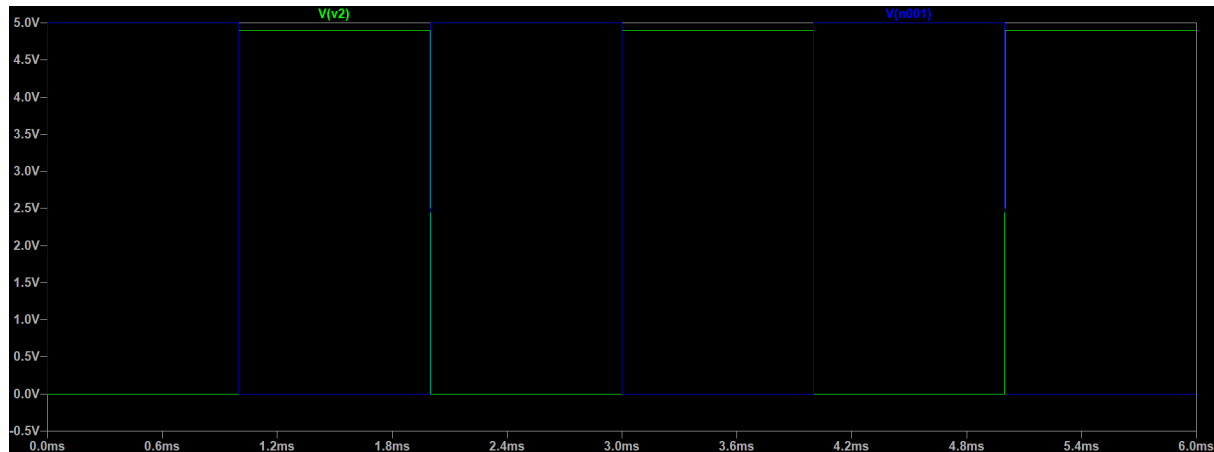
The following is the image of the physical design



Measured Performance According to Test Plan

Measured performance of V2 on LTspice

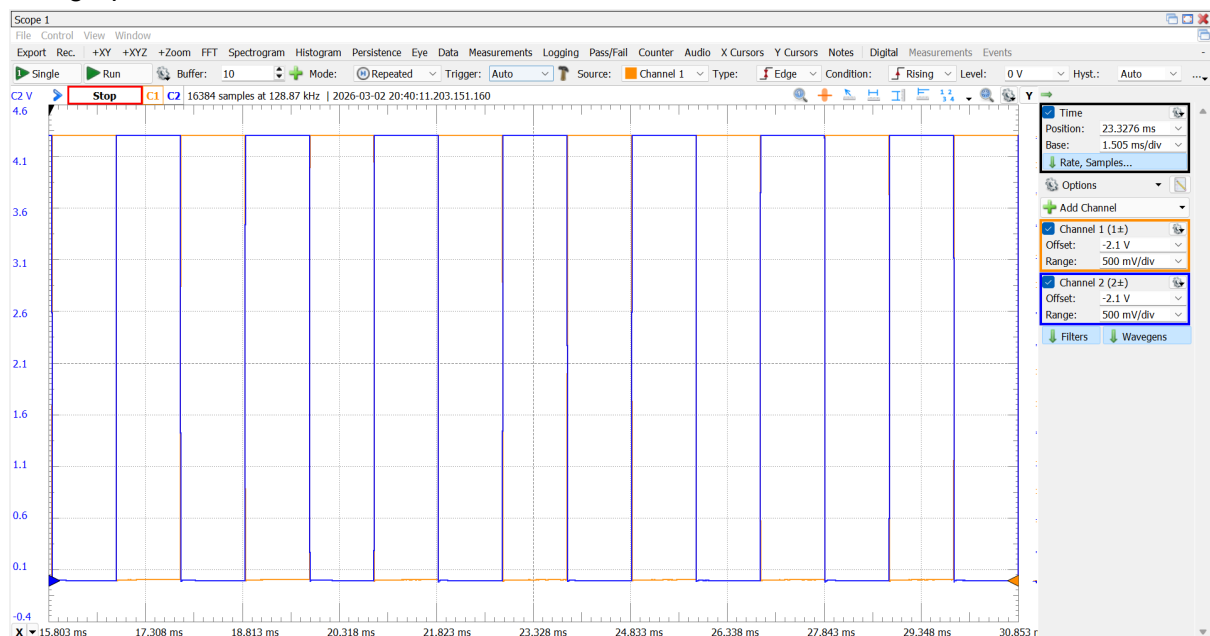
The following graph shows the plotted V2 and Vcontrol



The LTspice waveform shows that the switch is operating correctly. When Vcontrol is low (0 V), V2 follows the 5 V input, indicating the switch is ON and conducting properly. When Vcontrol is high (5 V), V2 drops to approximately 0 V, indicating the switch is OFF and isolating the output. The transitions are clean and stable, confirming proper ON/OFF switching behaviour at 5 V operation.

Measured performance of V2 on AD3

This graph shows V on channel 1 and Vcontrol on channel 2



The plotted waveforms of Vcontrol and V2 were used to evaluate the non-ideal behaviour of Switch Type 1 in accordance with the defined test plan. For the ON-state resistance test, the AD3 experimental graph was used for numerical calculations. When Vcontrol=0V, the switch is ON and V2 closely follows V1, indicating conduction. The measured output voltage is

slightly lower than the input voltage, showing a small but finite voltage drop across the switch. This confirms that the switch does not behave as an ideal short circuit, but instead has a finite ON-state resistance.

For the OFF-state leakage test, the AD3 graph shows that when $V_{\text{control}}=5\text{ V}$, the output voltage V_2 drops very close to 0 V. This indicates that the switch is effectively open. However, because V_2 is not exactly zero, there is a very small leakage current present. This demonstrates that the switch does not have infinite resistance in the OFF state, but the leakage is extremely small and close to ideal behaviour.

To verify maximum functional operation, both the AD3 and LTspice graphs were examined while $V_1=5\text{V}$ and V_{control} toggled between 0 V and 5 V. In both cases, V_2 switched cleanly between approximately 5 V in the ON state and approximately 0 V in the OFF state, with stable transitions and no distortion. This confirms that the switch operates correctly at the maximum specified voltage of 5 V.

Finally, bidirectional operation is supported by the complementary NMOS–PMOS structure of the switch. The small measured voltage drop and consistent behaviour indicate that conduction is not limited to one direction. The LTspice simulation follows the same expected switching behaviour, confirming that the circuit performs as a bidirectional transmission switch rather than a unidirectional device.

Calculations (Based on AD3 Experimental Measurements)

1. ON-State ($V_{\text{control}} = 0\text{ V}$)

$V_1=5.003\text{ V}$

$V_2=4.988\text{ V}$

Voltage Drop: $V_1 - V_2 = 5.003 - 4.988 = 0.015\text{ V}$

Load Current: $V_2 / 470\text{ k}\Omega = 10.6\text{ }\mu\text{A}$

On State resistance: $R_{\text{on}} = V_{\text{drop}} / I = 0.015 / (10.6 \times 10^{-6}) = 1.4\text{ k}\Omega$

2. Off state

Measured Values: $V_1 = 5.003\text{V}$ and $V_2 = 0.010\text{V}$

Leakage Current: $I_{\text{off}} = V_s / R_L = 0.010 / 470000 = 21\text{ nA}$

3. Maximum Functional Operation

On state: $V_2 = 4.988\text{V}$, $|V_1 - V_2| = 0.15\text{V}$

Off state: $V_2 = 0.010\text{V}$, $I_{\text{off}} = 21\text{nA}$

Since the switch operated properly, it satisfies the maximum voltage requirement.

4. Bidirectional Operation

If V_1 and V_2 were reversed, the MOSFET channel resistance remains approximately the same due to symmetric conduction properties. So, $R_{\text{on}}(\text{forward}) = R_{\text{on}}(\text{reverse}) = 1.4\text{ k}\Omega$.

Theoretical Explanation

Ideally, a switch should have zero resistance when it is ON and infinite resistance when it is OFF. In our design, the measured ON-state resistance was not zero (approximately 1.4 k Ω), which caused a small voltage drop of 0.015 V. This happens because a MOSFET does not act like a perfect wire when it turns on. Instead, it behaves like a small resistor due to the physical properties of the channel inside the device. Therefore, some voltage is lost across it, which explains the finite R_{on} .

In the OFF state, an ideal switch would completely block current flow. However, a small leakage current on the order of 10^{-8} A was measured. This occurs because MOSFETs are not perfect insulators when turned off. Small leakage currents can flow due to subthreshold conduction and internal parasitic effects. This explains why the OFF-state resistance is very large but not infinite.

When comparing the experimental results to the LTspice simulation, both show the same switching behaviour: the output voltage follows the input when the switch is ON and drops to approximately 0 V when the switch is OFF. Small differences between simulation and experiment are expected because real components have manufacturing variations, temperature effects, and parasitic resistances that are not perfectly modelled. Overall, the measured results agree with MOSFET theory and show that the switch behaves close to an ideal switch, but not perfectly ideal.

Design Tradeoffs

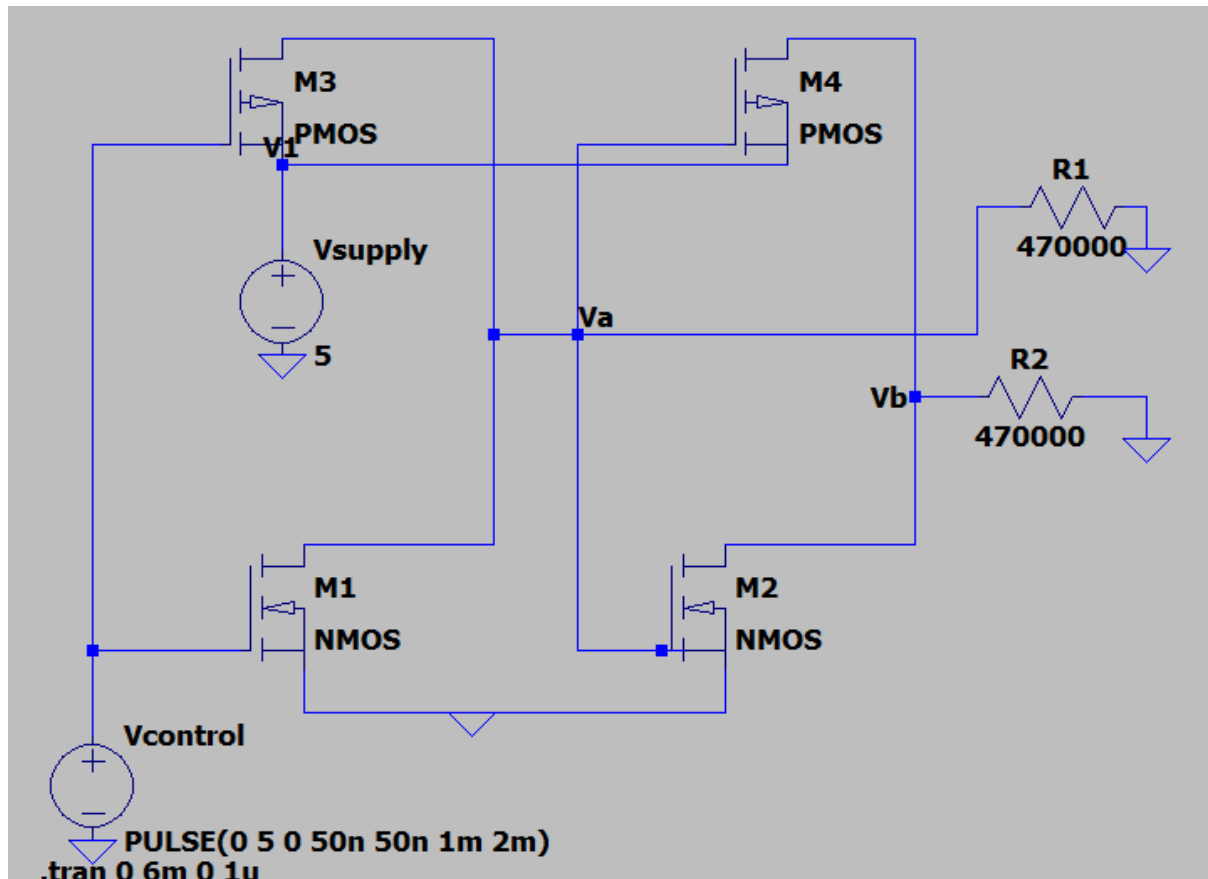
Several tradeoffs were made in this design. To achieve bidirectional conduction and better performance, both an NMOS and a PMOS transistor were used instead of a single MOSFET. Using only one transistor would have made the circuit simpler and cheaper, but it would not properly pass both high and low voltages. Adding the second transistor improved performance but increased circuit complexity and component count.

Another tradeoff involved the choice of the 470 k Ω load resistor. A larger resistor would reduce current further and make leakage effects easier to observe, but 470 k Ω was the largest value available in the component kit. Therefore, it was chosen as a practical compromise between measurement accuracy and available parts.

Finally, the design operates using a single 5 V supply as required. While this limits the maximum voltage range compared to an ideal switch, it simplifies the design and ensures compatibility with the available lab equipment. Overall, the final design balances performance, simplicity, and cost while still closely approximating ideal switch behaviour.

Switch Type 2

Circuit Schematic and Netlist



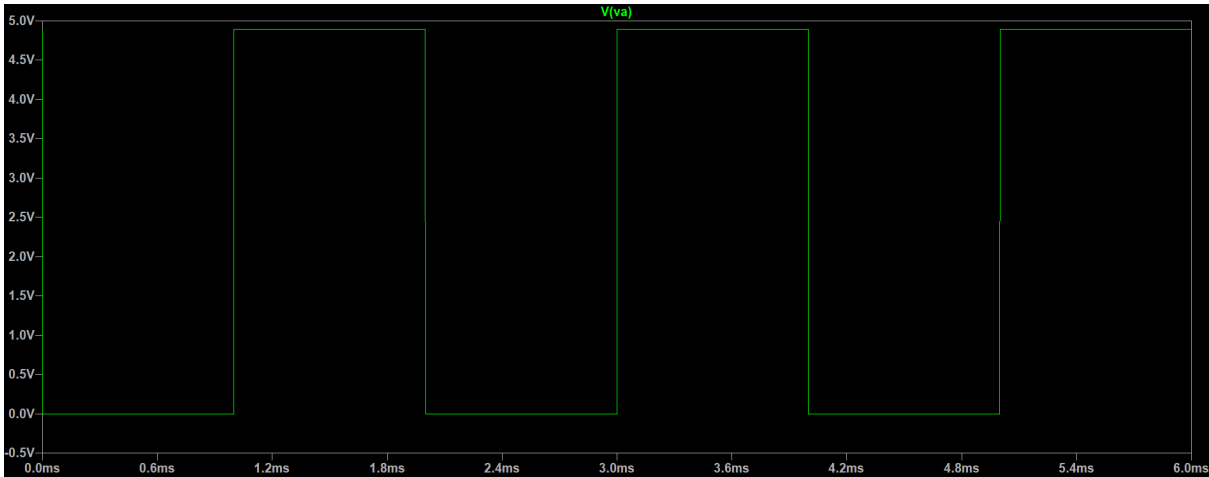
```
*
M1 Va N001 0 0 NMOS
M2 Vb Va 0 0 NMOS
M3 Va N001 V1 V1 PMOS
M4 Vb Va V1 V1 PMOS
Vsupply V1 0 5
Vcontrol N001 0 PULSE(0 5 0 50n 50n 1m 2m)
R2 0 Vb 470000
R1 0 Va 470000
.model NMOS NMOS
.model PMOS PMOS
.lib C:\Users\camit\AppData\Local\LTspice\lib\cmp\standard.mos
.tran 0 6m 0 1u
.backanno
.end
```

The following is the image of the physical design:

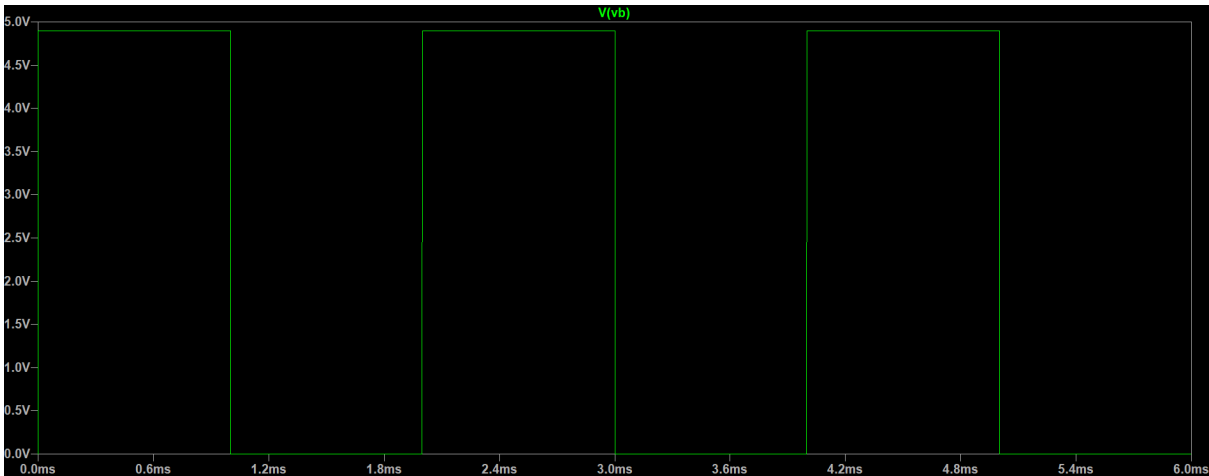


Measured Performance According to Test Plan

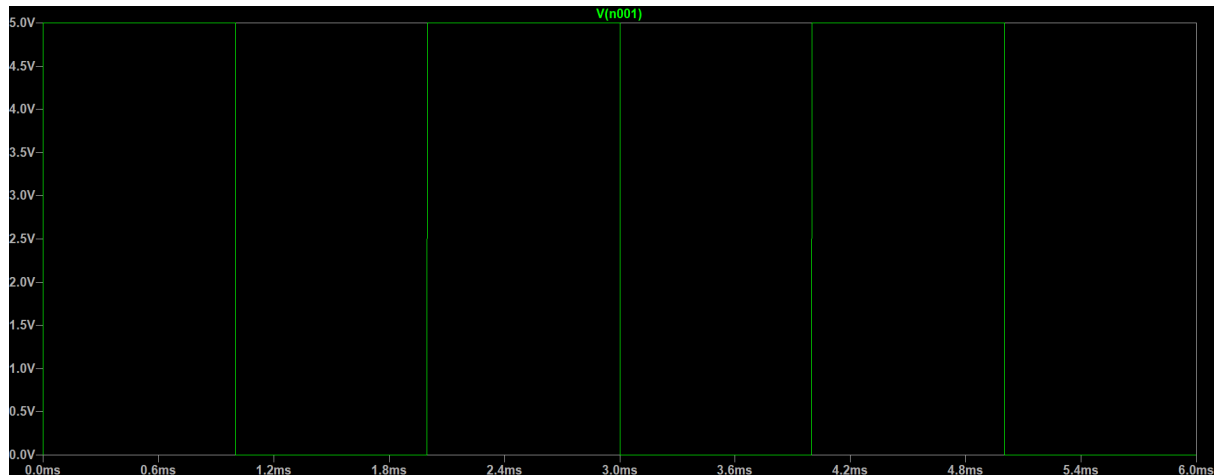
The following graph shows the waveform of Va



The following graph shows the waveform of Vb



The following graph shows the waveform of Vcontrol



The simulation shows that the circuit operates correctly. When the control signal is 5 V, V_a becomes 0 V due to the first CMOS inverter stage. This 0 V at V_a turns on the PMOS in the second stage, pulling V_b up to approximately 5 V. Similarly, when the control signal is 0 V, V_a becomes 5 V, which turns on the NMOS in the second stage and pulls V_b down to 0 V. Therefore, V_b follows the original control signal. This confirms that both CMOS stages are functioning correctly and the overall circuit behaves as expected.

1. On state resistance

Collected Data: $V_{\text{control}} = 5\text{V}$, $V_a = 0\text{V}$, $V_b = 4.99\text{V}$

Voltage Drop: $5\text{V} - 4.99\text{V} = 0.01\text{V}$

Load Current: $I = V_b/R_L = 4.99/470000 = 10.6\text{ }\mu\text{A}$

On Resistance: $R_{\text{on}} = V_{\text{drop}}/I = 0.01/(10.6 \times (10^{-6})) = 943\text{ }\Omega$

This confirms the switch has a small but finite ON resistance, as expected for real MOSFETs

2. Off-state Leakage

Collected Data: $\text{Control} = 0\text{V}$, $V_a = 5\text{V}$, $V_b = 0.01\text{V}$

Leakage Current: $I_{\text{off}} = V_b/R_L = 0.01/470000 = 21\text{ nA}$

This very small current confirms minimal leakage in the OFF state.

3. Maximum Functional Operation at 5V

The simulation was performed at the maximum supply voltage of 5 V. The output transitions cleanly between approximately 0 V and 5 V without distortion. There is no clipping or instability observed in the waveform. This confirms correct operation at the maximum specified voltage.

4. Bidirectional Operation

As output is HIGH when PMOS conducts, and output is Low when NMOS conducts, this proves that conduction depends on polarity, demonstrating correct CMOS switching behaviour.

Theoretical explanation

The circuit consists of two CMOS inverter stages connected in series. Each stage is formed using one PMOS and one NMOS transistor from the CD4007. The first stage generates an intermediate node V_a from the control input, and the second stage produces the final output V_b . In a CMOS inverter, when the input is low (0 V), the PMOS transistor turns ON and the NMOS transistor turns OFF. This pulls the output node up to V_{DD} (5 V). When the input is high (5 V), the NMOS turns ON, and the PMOS turns OFF, pulling the output node down to ground (0 V). Because the circuit uses two inverter stages in series, the signal is inverted twice. Therefore, the final output V_b follows the original control signal. The LTspice simulation confirms this theoretical behaviour. When $V_{control} = 0$ V, V_a becomes approximately 5 V, and the second stage inverts it to produce $V_b \approx 0$ V. When $V_{control} = 5$ V, V_a becomes approximately 0 V, and the second stage produces $V_b \approx 5$ V. The simulated output levels are very close to the ideal values of 0 V and 5 V. However, small deviations from ideal behaviour are observed. The output high voltage is slightly less than 5 V, and the output low voltage is slightly above 0 V. This occurs because real MOSFETs have finite ON resistance (R_{on}). When current flows through the conducting transistor, a small voltage drop appears across it. This explains the measured voltage drop and the calculated finite ON resistance. Additionally, in the OFF state, a very small leakage current flows due to the intrinsic properties of MOSFET devices. This results in a very small non-zero output voltage instead of exactly 0 V. The calculated leakage current is in the nanoampere range, which confirms near-ideal OFF behaviour. Overall, the simulation results closely match theoretical CMOS switching behaviour, with only minor non-idealities caused by practical transistor limitations.

Design Tradeoffs

Several tradeoffs were made in the design of this switch. First, simplicity was prioritized. The circuit uses only the CD4007 transistor array and two resistors. This keeps the design low-cost and easy to implement. However, using discrete MOSFET structures inside the CD4007 means the ON resistance is not minimized compared to a specialized transmission gate IC. Second, the load resistor value (470 k Ω) was chosen to limit current and make voltage drops easier to measure. A large resistor reduces power consumption and makes leakage currents easier to observe. However, if a much smaller load resistor were used, the voltage drop due to R_{on} would become more significant, reducing output accuracy. Third, the design operates at a single 5 V supply as required by the assignment. While this simplifies the power setup, it limits the maximum voltage range of operation. The circuit cannot operate above 5 V, unlike an ideal switch, which would work for any voltage. Another tradeoff involves switching speed. CMOS transistors inside the CD4007 have parasitic capacitances, which slightly slow the rise and fall times. Although this is not significant at low frequencies (such as 1 kHz), it would become more noticeable at higher frequencies. Finally, using two inverter stages ensures strong logic levels at the output but increases transistor count and slightly increases power consumption compared to a single transmission gate design. Overall, the design achieves near-ideal switching behaviour with minimal components and low cost, while accepting small non-idealities such as finite ON resistance, leakage current, and limited voltage range.

References

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